

TECHNICAL REPORT · JUNE 2026

# AdalahCredit A-Z Report

Unified AI API Gateway & Marketplace — Full architecture, code structure, flow analysis, and feature documentation.

45K

LINES OF CODE

197

SOURCE FILES

321+

AI MODELS

32

DB TABLES

v1.0 · Generated automatically

# Table of Contents

---

- 1. Executive Summary
- 2. Tech Stack
- 3. Logic Structure — Architecture
- 4. Code Structure — File Organization
- 5. Flow Structure — User & API Flows
- 6. Database Schema — 32 Tables
- 7. API Endpoints — 21 Routes
- 8. Proxy Engine — Request Pipeline
- 9. Subscription & Billing
- 10. OAuth Provider System — 17 Providers
- 11. Frontend — 26 Pages
- 12. Security & Bug Fixes
- 13. Infrastructure & Deployment
- 14. Complete Feature Checklist
- 15. Project Statistics

# 1. Executive Summary

---

**AdalahCredit** is a full-stack AI API gateway and marketplace platform that provides unified access to 300+ AI models from multiple providers through a single OpenAI-compatible API. Users get one API key to access models from Groq, NVIDIA, OpenAI, Anthropic, Google, and many others.

45,021

LINES OF CODE

197

SOURCE FILES

32

DATABASE  
TABLES

26

PAGES

## Key Capabilities

| CAPABILITY          | DESCRIPTION                                              |
|---------------------|----------------------------------------------------------|
| Unified API         | Single OpenAI-compatible endpoint for all providers      |
| Multi-Provider      | 204 providers seeded, 321+ models, 6 actively configured |
| 4-Tier Fallback     | Admin key → OAuth → User connections → Free tier         |
| 5 Proxy Endpoints   | Chat, embeddings, images, audio (TTS/STT), moderations   |
| 17 OAuth Providers  | Claude, Codex, OpenAI, Gemini, GitHub, Cursor, etc.      |
| Subscription Engine | 4-tier plans (Free/Starter/Pro/Ultime) with PAYG credits |
| Admin Dashboard     | 9 tabs — full system management                          |
| User Dashboard      | 10 tabs — keys, usage, billing, team, webhooks           |
| i18n                | English + Bahasa Indonesia, auto-detect                  |
| Dark/Light Mode     | System-aware with manual toggle                          |

## 2. Tech Stack

---

### Frontend

| TECHNOLOGY    | VERSION | PURPOSE                            |
|---------------|---------|------------------------------------|
| React         | 19      | UI framework                       |
| Vite          | 6       | Build tool & dev server            |
| TypeScript    | 5+      | Type safety                        |
| Tailwind CSS  | v4      | Utility-first styling              |
| shadcn/ui     | latest  | Component library (60+ components) |
| Framer Motion | latest  | Animations & transitions           |
| Recharts      | latest  | Analytics charts                   |
| React Router  | v6      | Client-side routing                |

### Backend

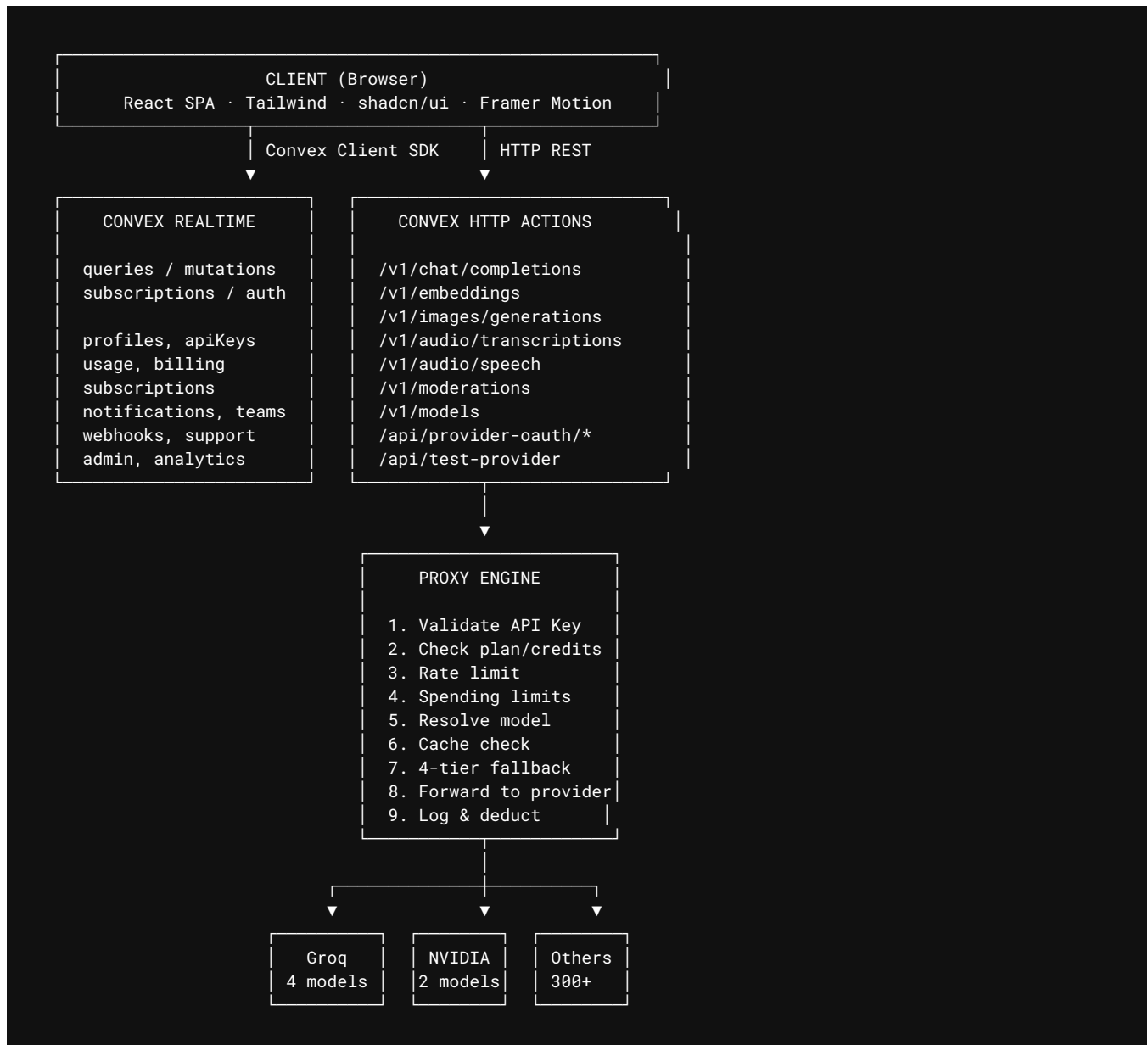
| TECHNOLOGY   | PURPOSE                                                 |
|--------------|---------------------------------------------------------|
| Convex       | Real-time serverless backend with database              |
| Convex Auth  | Authentication (email/password + OAuth)                 |
| HTTP Actions | RESTful API proxy endpoints (36 routes)                 |
| Cron Jobs    | 7 background tasks (cache cleanup, health checks, etc.) |

# Infrastructure

| TECHNOLOGY    | PURPOSE                       |
|---------------|-------------------------------|
| Viktor Spaces | Frontend hosting & deployment |
| Convex Cloud  | Backend-as-a-Service          |
| Bun           | Package manager & runtime     |
| GitHub        | Source control                |

# 3. Logic Structure

## 3.1 High-Level Architecture



## 3.2 Proxy Request Pipeline

01

### Validate Key

Check API key exists, not revoked, check IP allowlist

02

### Resolve Model

DB lookup → static map fallback → real provider model ID mapping

03

### Access Check

Plan tier, credit balance, subscription expiry, rate limit, spending cap

04

### Cache Lookup

SHA-256 hash → check responseCache table → return if HIT

05

### Build Request

Provider-specific builders (OpenAI, Anthropic, Google, Cohere)

06

### 4-Tier Fallback

Admin key → OAuth → User connections → Free tier; auto-retry on 429/5xx

07

### Forward

Send to provider, stream SSE or buffer non-streaming response

08

### Post-Process

Log usage, deduct credits, cache response, dispatch webhooks

## 3.3 Subscription & Billing Logic

| FEATURE         | FREE      | STARTER (RP 50K) | PRO (RP 200K) | ULTIMATE (RP 500K) |
|-----------------|-----------|------------------|---------------|--------------------|
| Monthly Credits | \$1.00    | \$10.00          | \$50.00       | \$200.00           |
| API Keys        | 2         | 5                | 20            | Unlimited          |
| Rate Limit      | 10 RPM    | 30 RPM           | 60 RPM        | 120 RPM            |
| Daily Requests  | 200       | 2,000            | 10,000        | Unlimited          |
| Model Access    | Free only | Free + Cheap     | + Standard    | All tiers          |

# 4. Code Structure

---

## 4.1 Backend — 62 files, 19,587 lines

### Core Proxy (5 files, ~2,781 lines)

| FILE                | LINES | PURPOSE                                                 |
|---------------------|-------|---------------------------------------------------------|
| proxy.ts            | 927   | Main chat completions handler, model mapping, streaming |
| proxyInternal.ts    | 533   | Validate key, rate limit, cache, cost calculation       |
| proxyConnections.ts | 816   | Connection pool, fallback management, health tracking   |
| providerAuth.ts     | 346   | Model resolution, provider authentication               |
| providerMeta.ts     | 265   | Provider metadata & capabilities                        |

### Additional Proxy Endpoints (4 files)

| FILE                | ENDPOINT                                    |
|---------------------|---------------------------------------------|
| proxyEmbeddings.ts  | /v1/embeddings                              |
| proxyImages.ts      | /v1/images/generations                      |
| proxyAudio.ts       | /v1/audio/transcriptions & /v1/audio/speech |
| proxyModerations.ts | /v1/moderations                             |



## Auth & Users (6 files)

| FILE                                               | PURPOSE                                               |
|----------------------------------------------------|-------------------------------------------------------|
| <code>auth.ts</code> / <code>auth.config.ts</code> | Convex Auth configuration                             |
| <code>profiles.ts</code> (211 lines)               | User profiles CRUD, onboarding, referral codes        |
| <code>users.ts</code>                              | Account deletion with full cascade cleanup            |
| <code>apiKeys.ts</code>                            | API key CRUD with <code>crypto.getRandomValues</code> |
| <code>testAuth.ts</code>                           | Test user authentication for development              |

## OAuth System (5 files, ~2,602 lines)

| FILE                                | LINES | PURPOSE                                        |
|-------------------------------------|-------|------------------------------------------------|
| <code>oauthConstants.ts</code>      | 317   | 17 OAuth provider configurations               |
| <code>oauthActions.ts</code>        | 400   | OAuth flow actions (start, callback, complete) |
| <code>oauthProviders.ts</code>      | 1,034 | Provider management & token operations         |
| <code>providerOAuth.ts</code>       | 472   | OAuth token management & refresh               |
| <code>providerConnections.ts</code> | 379   | Connection CRUD & upsert                       |

## Billing & Subscriptions (4 files, ~2,622 lines)

| FILE                               | LINES | PURPOSE                                            |
|------------------------------------|-------|----------------------------------------------------|
| <code>subscriptionEngine.ts</code> | 941   | Full subscription & PAYG engine, plan management   |
| <code>billing.ts</code>            | —     | Billing queries & mutations                        |
| <code>payments.ts</code>           | 377   | Payment orders, proof upload, admin confirm/reject |
| <code>pricing.ts</code>            | 304   | Pricing config, markup, model cost calculation     |

## Infrastructure (4 files)

| FILE                      | LINES | PURPOSE                            |
|---------------------------|-------|------------------------------------|
| <code>schema.ts</code>    | 639   | 32 tables, 64 indexes              |
| <code>http.ts</code>      | 294   | HTTP router with 36 route bindings |
| <code>cronTasks.ts</code> | 356   | 7 cron task implementations        |
| <code>crons.ts</code>     | —     | Cron schedule definitions          |

## 4.2 Frontend — 135 files, 25,064 lines

### Top 10 Largest Frontend Files

| FILE                               | LINES | PURPOSE                         |
|------------------------------------|-------|---------------------------------|
| <code>AdminDashboard.tsx</code>    | 2,058 | Admin dashboard — 9 tabs        |
| <code>UserDashboard.tsx</code>     | 1,621 | User dashboard — 10 tabs        |
| <code>PublicLandingPage.tsx</code> | 1,131 | Marketing landing page          |
| <code>DocsPage.tsx</code>          | 1,035 | API documentation               |
| <code>ModelsPage.tsx</code>        | 1,029 | Model catalog browser           |
| <code>PlaygroundPage.tsx</code>    | 868   | Interactive API playground      |
| <code>i18n.tsx</code>              | 631   | Internationalization (EN + ID)  |
| <code>ConnectionsPanel.tsx</code>  | 586   | Connection management UI        |
| <code>TopUpPage.tsx</code>         | 537   | Credit purchase / plan upgrade  |
| <code>ConnectionRow.tsx</code>     | 509   | Individual connection component |

## 5. Flow Structure

---

### 5.1 User Registration → First API Call

01

#### Signup

Email + password registration via Convex Auth

02

#### Onboarding

Create profile, set name & company, auto-assign Free plan

03

#### Create API Key

Dashboard → Keys tab → generate `sk-ac-xxx` key

04

#### First Request

Use key in Playground or curl to call `/v1/chat/completions`

### 5.2 OAuth Provider Connection

01

#### Choose Provider

Select from 17 providers in Connections tab

02

#### PKCE Auth

Generate challenge, open provider popup, user authorizes

03

#### Token Exchange

Exchange code for access + refresh tokens, store encrypted

04

#### Auto Refresh

Cron job refreshes tokens every 30 minutes

## 5.3 Payment Flow

01

### Create Order

User selects plan/top-up amount, creates payment order

02

### Bank Transfer

User transfers to bank account (BCA, Mandiri, etc.)

03

### Upload Proof

Upload payment screenshot, rate-limited to 1/min

04

### Admin Review

Admin confirms → credits added; or rejects with reason

## 5.4 4-Tier Fallback Chain

| TIER | SOURCE                    | PRIORITY | WHEN USED                                   |
|------|---------------------------|----------|---------------------------------------------|
| 1    | Admin provider API key    | Highest  | First attempt — fastest, most reliable      |
| 2    | OAuth connections         | High     | Fallback if Tier 1 fails (429/5xx)          |
| 3    | User provider connections | Medium   | Fallback if Tier 2 exhausted                |
| 4    | Free-tier providers       | Lowest   | Last resort — e.g., NVIDIA NIM free credits |

# 6. Database Schema

---

32 tables with 64 indexes in Convex, organized by domain.

## Core Tables

| TABLE         | PURPOSE                | KEY FIELDS                                          |
|---------------|------------------------|-----------------------------------------------------|
| profiles      | User profiles          | userId, email, name, role, plan, referralCode       |
| apiKeys       | API keys (hashed)      | userId, key, name, status, expiresAt, allowedIPs    |
| usageLogs     | Per-request logs       | apiKeyId, model, provider, tokens, cost, latencyMs  |
| dailyUsage    | Aggregated daily stats | userId, date, totalRequests, totalTokens, totalCost |
| responseCache | Response caching       | requestHash, responseJson, expiresAt, hitCount      |

## Billing Tables

| TABLE              | PURPOSE                                    |
|--------------------|--------------------------------------------|
| subscriptions      | Active subscriptions with credit balance   |
| subscriptionOrders | Subscription purchase orders               |
| subscriptionPlans  | Plan definitions (price, limits, features) |
| transactions       | Credit transactions history                |
| paymentOrders      | Payment orders with proof uploads          |
| paymentGateways    | Payment method configurations              |
| pricingConfig      | Global markup & pricing settings           |

## Provider Tables

| TABLE                            | PURPOSE                                      |
|----------------------------------|----------------------------------------------|
| <code>providerConfigs</code>     | Provider API keys, models, base URLs         |
| <code>providerConnections</code> | OAuth/API key connections with health status |
| <code>providerOAuthStates</code> | Temporary OAuth flow state storage           |
| <code>routingState</code>        | Connection routing & failover state          |

## Feature Tables

| TABLE                                                  | PURPOSE                               |
|--------------------------------------------------------|---------------------------------------|
| <code>teams</code> / <code>teamMembers</code>          | Team organizations & membership       |
| <code>webhooks</code> / <code>webhookDeliveries</code> | User webhooks & delivery logs         |
| <code>notifications</code>                             | In-app notification system            |
| <code>referrals</code>                                 | Referral program tracking             |
| <code>modelAliases</code>                              | User-defined model name aliases       |
| <code>spendingAlerts</code>                            | Per-key spending limit configurations |
| <code>supportTickets</code>                            | Customer support tickets              |
| <code>blogPosts</code>                                 | Blog CMS content                      |
| <code>changelogEntries</code>                          | Product changelog                     |
| <code>serviceStatus</code> / <code>incidents</code>    | Service health & incident tracking    |
| <code>auditLogs</code>                                 | Admin action audit trail              |
| <code>errorLogs</code>                                 | System error tracking                 |

## 7. API Endpoints

---

### Proxy API (OpenAI-Compatible)

| METHOD | PATH                     | DESCRIPTION                                  |
|--------|--------------------------|----------------------------------------------|
| POST   | /v1/chat/completions     | Chat completions (streaming & non-streaming) |
| GET    | /v1/models               | List available models                        |
| POST   | /v1/embeddings           | Text embeddings                              |
| POST   | /v1/images/generations   | Image generation (DALL-E compatible)         |
| POST   | /v1/audio/transcriptions | Speech-to-text (Whisper compatible)          |
| POST   | /v1/audio/speech         | Text-to-speech                               |
| POST   | /v1/moderations          | Content moderation                           |

### OAuth & Provider API

| METHOD | PATH                                  | DESCRIPTION                    |
|--------|---------------------------------------|--------------------------------|
| POST   | /api/provider-oauth-v2/authorize      | Start PKCE OAuth flow          |
| POST   | /api/provider-oauth-v2/exchange       | Exchange authorization code    |
| POST   | /api/provider-oauth-v2/device-code    | Start device code flow         |
| POST   | /api/provider-oauth-v2/poll           | Poll device code status        |
| POST   | /api/provider-oauth-v2/import-token   | Import existing token          |
| GET    | /api/provider-oauth-v2/list-providers | List available OAuth providers |
| POST   | /api/test-provider                    | Test provider API key          |
| POST   | /api/test-connection                  | Test connection health         |

# 8. Proxy Engine Details

## Model ID Mapping

Internal model IDs are mapped to real provider API names:

| INTERNAL ID        | PROVIDER API ID                           | PROVIDER |
|--------------------|-------------------------------------------|----------|
| groq/llama-4-scout | meta-llama/llama-4-scout-17b-16e-instruct | Groq     |
| groq/llama-3.3-70b | llama-3.3-70b-versatile                   | Groq     |
| groq/qwen3-32b     | qwen/qwen3-32b                            | Groq     |
| groq/llama-3.1-8b  | llama-3.1-8b-instant                      | Groq     |
| nv/llama-3.3-70b   | meta/llama-3.3-70b-instruct               | NVIDIA   |
| nv/mistral-large   | mistralai/mistral-large-2-instruct        | NVIDIA   |

## Provider Request Builders

| BUILDER                  | PROVIDERS                                    | FORMAT                     |
|--------------------------|----------------------------------------------|----------------------------|
| buildOpenAICompatRequest | Groq, NVIDIA, OpenAI, Mistral, xAI, DeepSeek | OpenAI API format          |
| buildAnthropicRequest    | Anthropic (Claude)                           | Anthropic Messages API     |
| buildGoogleRequest       | Google (Gemini)                              | Gemini generateContent API |
| buildCohereRequest       | Cohere                                       | Cohere Chat API v2         |



## Cron Jobs

| JOB               | SCHEDULE     | PURPOSE                                   |
|-------------------|--------------|-------------------------------------------|
| Cache cleanup     | Every 10 min | Remove expired response cache entries     |
| Key expiration    | Every 1 hour | Revoke expired API keys                   |
| Daily aggregation | 00:05 UTC    | Roll up per-request logs into daily stats |
| Error log cleanup | 00:30 UTC    | Delete error logs older than 30 days      |
| Health check      | Every 5 min  | Ping providers, update service status     |
| OAuth refresh     | Every 30 min | Proactively refresh expiring tokens       |
| Cooldown cleanup  | Every 5 min  | Clear expired rate-limit cooldowns        |

# 9. Subscription & Billing

## Model Tier Classification

| TIER     | EXAMPLES                             | AVAILABLE TO |
|----------|--------------------------------------|--------------|
| FREE     | NVIDIA NIM models (free API credits) | All plans    |
| CHEAP    | Groq models (ultra-fast, low cost)   | Starter+     |
| STANDARD | GPT-4o-mini, Claude Haiku            | Pro+         |
| PREMIUM  | GPT-4o, Claude Opus, Gemini Pro      | Ultimate     |

## Cost Calculation Formula

```
cost_usd = (input_tokens / 1,000,000) × input_price_per_1M  
          + (output_tokens / 1,000,000) × output_price_per_1M
```

```
cost_with_markup = cost_usd × (1 + markup_percent / 100)  
cost_cents = cost_with_markup × 100
```

→ Deducted from user's credit balance after each request

# 10. OAuth Provider System

---

## 17 Supported Providers

| #  | PROVIDER           | SLUG        | PKCE | DEVICE CODE | IMPORT |
|----|--------------------|-------------|------|-------------|--------|
| 1  | Claude (Anthropic) | claude      | ✓    | —           | ✓      |
| 2  | Codex (OpenAI)     | codex       | ✓    | —           | —      |
| 3  | OpenAI Native      | openai      | ✓    | —           | —      |
| 4  | Gemini CLI         | gemini-cli  | ✓    | —           | —      |
| 5  | GitHub Copilot     | github      | ✓    | ✓           | —      |
| 6  | Kiro (AWS)         | kiro        | ✓    | ✓           | —      |
| 7  | GitLab Duo         | gitlab-duo  | ✓    | ✓           | —      |
| 8  | Cursor             | cursor      | —    | ✓           | ✓      |
| 9  | Windsurf           | windsurf    | —    | ✓           | ✓      |
| 10 | Trae               | trae        | —    | ✓           | ✓      |
| 11 | Qwen               | qwen        | ✓    | —           | —      |
| 12 | Kimi Coding        | kimi-coding | ✓    | —           | —      |
| 13 | Cline              | cline       | ✓    | —           | —      |
| 14 | Kilocode           | kilocode    | ✓    | —           | —      |
| 15 | Antigravity        | antigravity | ✓    | —           | —      |
| 16 | AGY                | agy         | ✓    | —           | —      |
| 17 | Qoder              | qoder       | ✓    | —           | —      |

# 11. Frontend — 26 Pages

---

## Public Pages (14)

| ROUTE       | PAGE          | DESCRIPTION                                |
|-------------|---------------|--------------------------------------------|
| /           | Landing       | Marketing page with features, pricing, CTA |
| /docs       | Documentation | Full API docs with code examples           |
| /models     | Models        | Browse 321+ AI models with filters         |
| /compare    | Compare       | Side-by-side model comparison              |
| /blog       | Blog          | Blog article listing                       |
| /blog/:slug | Blog Post     | Individual article                         |
| /status     | Status        | Real-time service health                   |
| /changelog  | Changelog     | Product version history                    |
| /sdks       | SDKs          | SDK & library downloads                    |
| /support    | Support       | Submit support tickets                     |
| /terms      | Terms         | Terms of Service                           |
| /privacy    | Privacy       | Privacy Policy                             |
| /login      | Login         | Sign in form                               |
| /signup     | Signup        | Registration form                          |

## Authenticated Pages (12)

| ROUTE           | PAGE            | TABS / FEATURES                                                                                 |
|-----------------|-----------------|-------------------------------------------------------------------------------------------------|
| /dashboard      | User Dashboard  | 10 tabs: overview, keys, usage, logs, billing, aliases, limits, settings, team, webhooks        |
| /admin          | Admin Dashboard | 9 tabs: overview, users, models, provider-config, connections, pricing, payments, audit, system |
| /playground     | Playground      | Interactive API testing with live responses                                                     |
| /topup          | Top Up          | Purchase credits or upgrade plan                                                                |
| /onboarding     | Onboarding      | New user setup wizard                                                                           |
| /referral       | Referral        | Referral program dashboard                                                                      |
| /receipt        | Receipt         | Payment receipt viewer                                                                          |
| /settings       | Settings        | Account settings (standalone)                                                                   |
| /oauth/callback | OAuth Callback  | OAuth redirect handler                                                                          |
| *               | 404             | Custom not found page                                                                           |

# 12. Security & Bug Fixes

## Security Measures

| AREA               | IMPLEMENTATION                                               |
|--------------------|--------------------------------------------------------------|
| API Keys           | <code>crypto.getRandomValues</code> , hashed before storage  |
| Authentication     | Convex Auth with JWT sessions                                |
| Admin Guard        | <code>requireAdmin()</code> on all admin mutations           |
| Ownership          | All user queries filter by authenticated <code>userId</code> |
| Rate Limiting      | Per-key RPM enforcement                                      |
| Spending Limits    | Per-key monthly caps with alerts                             |
| IP Allowlist       | Optional per-key IP restriction                              |
| OAuth              | PKCE + state parameter validation                            |
| Audit Trail        | All admin actions logged with timestamp & IP                 |
| Input Sanitization | Server-side validation on all mutations                      |
| Account Deletion   | Full cascade: keys, usage, subscriptions, team, webhooks     |

## 75 Bug Fixes Applied

| SEVERITY        | COUNT | EXAMPLES                                                                    |
|-----------------|-------|-----------------------------------------------------------------------------|
| <b>Critical</b> | 16    | Auth bypass, backdoor removal, ownership checks, internal mutation exposure |
| <b>High</b>     | 15    | Bounded queries, rate limiting, input validation                            |
| <b>Medium</b>   | 28    | Error handling, controlled inputs, state management                         |
| <b>Low</b>      | 16    | Pagination, negative value rejection, UI cleanup                            |

Fixes applied across 19 files: +494 lines added, -200 lines removed.

# 13. Infrastructure & Deployment

---

## Deployment Stack

| LAYER            | SERVICE             | URL                                                            |
|------------------|---------------------|----------------------------------------------------------------|
| Frontend Hosting | Viktor Spaces       | <code>preview-api-gateway-landing-45bd0329.viktor.space</code> |
| Backend          | Convex Cloud        | <code>blessed-sardine-879.convex.cloud</code>                  |
| API Proxy        | Convex HTTP Actions | <code>blessed-sardine-879.convex.site</code>                   |
| Source Control   | GitHub              | <code>github.com/fadilismee/adalahcredit-new</code>            |

## Deploy Process

```
# 1. Deploy Convex backend
CONVEX_DEPLOY_KEY=... bunx convex deploy --cmd 'echo skip'

# 2. Build frontend
bun run vite build

# 3. Deploy to Viktor Spaces
uv run python temp/deploy_v4.py
```

# 14. Complete Feature Checklist

---

## Core Platform

- ✓ Landing page with pricing & features
- ✓ User registration & login (email/password)
- ✓ User profile & onboarding flow
- ✓ Admin & user role separation
- ✓ Dark/light theme with system detection
- ✓ English & Indonesian i18n
- ✓ Custom 404 page & error boundaries

## API Gateway

- ✓ OpenAI-compatible chat completions (streaming + non-streaming)
- ✓ Model listing, embeddings, images, audio, moderations
- ✓ Response caching with TTL & hash deduplication
- ✓ Provider-specific request builders (4 formats)
- ✓ Model ID mapping (internal → provider)
- ✓ 4-tier fallback with auto-retry on 429/5xx

## Multi-Provider System

- ✓ 204 providers seeded, 321+ models cataloged
- ✓ Dynamic provider configuration via admin UI
- ✓ Connection health tracking & failover
- ✓ 17 OAuth providers (PKCE, device code, import)
- ✓ Automatic OAuth token refresh (cron)

## Billing & Subscriptions

- ✓ 4-tier subscription plans with credit system



- ✓ Pay-as-you-go with per-request cost calculation
- ✓ Manual payment (bank transfer + proof)
- ✓ Admin payment confirmation/rejection workflow

## Dashboards

- ✓ User dashboard: 10 tabs (keys, usage, billing, team, webhooks...)
- ✓ Admin dashboard: 9 tabs (users, models, pricing, payments, audit...)
- ✓ Interactive API playground
- ✓ Analytics charts with Recharts

## Platform Features

- ✓ Blog CMS, changelog management
- ✓ Service status page with incidents
- ✓ Support ticket system
- ✓ SDK & libraries page
- ✓ Model comparison tool
- ✓ Referral program
- ✓ Notification system (in-app bell)
- ✓ Webhook event dispatch

## 15. Project Statistics

---

**45,021**

TOTAL LINES

**62**

BACKEND FILES

**135**

FRONTEND FILES

**60+**

UI COMPONENTS

**32**

DB TABLES

**64**

DB INDEXES

**36**

HTTP ROUTES

**7**

CRON JOBS

**17**

OAUTH PROVIDERS

**204**

SEEDED  
PROVIDERS

**321+**

SEEDED MODELS

**75**

BUG FIXES

END OF REPORT · ADALAHCREDIT v1.0 · JUNE 2026